

## END TERM EXAMINATION

### THIRD SEMESTER [B.TECH] JANUARY 2024

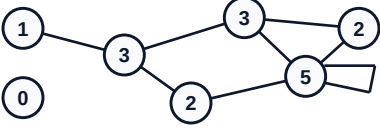
Paper Code: CIC-209

Subject: Data Structures

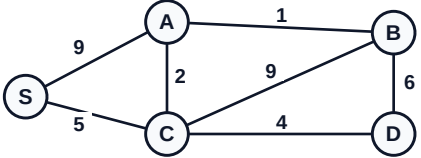
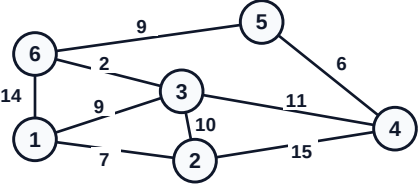
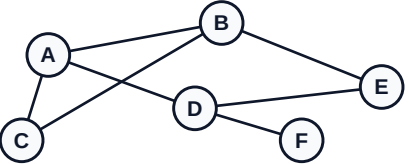
Time: 3 Hours

Maximum Marks: 75

**Note: Attempt five questions including Q.No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.**

| Q. No.           | Question                                                                                                                                                                                                       | Max. Marks | CO(s) |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|
| 1. (a)           | What do you understand by underflow and overflow in stack and queue data structures?                                                                                                                           | 3          | CO1   |
| (b)              | Why is it required to analyze the time complexity and space complexity of an algorithm?                                                                                                                        | 3          | CO1   |
| (c)              | Write short note on Bucket Hashing.                                                                                                                                                                            | 3          | CO2   |
| (d)              | Write an algorithm to count the total number of nodes in a binary tree.                                                                                                                                        | 3          | CO2   |
| (e)              | Represent the following graph using Adjacency Matrix and Adjacency List representations:                                      | 3          | CO2   |
| <b>UNIT – I</b>  |                                                                                                                                                                                                                |            |       |
| 2. (a)           | Convert the following infix expression into postfix expression using Stack:<br>$(A + B) / C * (D - E) ^ F - G$                                                                                                 | 5          | CO2   |
| (b)              | Write an algorithm to insert an element into a sorted doubly linked list.                                                                                                                                      | 5          | CO2   |
| (c)              | Write a short note on Double-Ended Queue (Deque). Can we implement stack and queue data structures using deque? Comment with justification.                                                                    | 5          | CO2   |
| 3. (a)           | What are the different classifications of data structures? Explain briefly all types of primitive and non-primitive data structures.                                                                           | 5          | CO1   |
| (b)              | Consider a 2D array of integers $A[-11..10, 9..20]$ . Each element requires 4 bytes. If the base address of Array A is 100, derive the address formula and determine the exact memory location of $A[5][15]$ . | 5          | CO2   |
| (c)              | Explain similarities and dissimilarities between stack and queue data structures. List two distinct practical applications of each.                                                                            | 5          | CO1   |
| <b>UNIT – II</b> |                                                                                                                                                                                                                |            |       |
| 4. (a)           | Step-by-step construct the binary tree based on the given traversals:<br><b>Preorder:</b> A B D C E G F H I <b>Inorder:</b> B D A G E C H F I                                                                  | 5          | CO2   |
| (b)              | What is an AVL Tree? Explain different types of rotations (LL, RR, LR, RL) used in AVL tree balancing with examples.                                                                                           | 5          | CO2   |
| (c)              | Write a short note on B-Tree properties, node structure, and search mechanics.                                                                                                                                 | 5          | CO2   |
| 5. (a)           | Insert the following elements sequentially into a Binary Search Tree (BST) and show the tree after each step:<br><b>B, D, A, G, E, C, H, F, I</b>                                                              | 5          | CO2   |
| (b)              | What is a sparse matrix? Explain its 3-tuple representation and give the node structure for a linked-list-based 3-tuple sparse matrix representation.                                                          | 5          | CO2   |
| (c)              | Insert the following elements into a Max-Heap tree sequentially and show the heap tree after each insertion:<br><b>20, 40, 10, 70, 50, 30, 80, 60, 90</b>                                                      | 5          | CO2   |

[P.T.O.]

| Q. No.            | Question                                                                                                                                                                                                                                        | Max. Marks | CO(s) |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|
| <b>UNIT – III</b> |                                                                                                                                                                                                                                                 |            |       |
| 6. (a)            | Write an algorithm for Merge Sort. Sort the following list of numbers using Merge Sort showing all recursion levels:<br><b>60, 2, 15, 20, 12, 115, 90, 50, 44, 85</b>                                                                           | 10         | CO2   |
| (b)               | Explain various collision resolution techniques used in hashing (Separate Chaining, Linear Probing, Quadratic Probing, Double Hashing).                                                                                                         | 5          | CO2   |
| 7. (a)            | What is hashing? What are the characteristics of a good hashing function? Explain various hash function methods (Division, Mid-square, Folding) with examples.                                                                                  | 10         | CO2   |
| (b)               | Differentiate between list search, sequential search, and binary search. Give the time complexity and an example of each.                                                                                                                       | 5          | CO2   |
| <b>UNIT – IV</b>  |                                                                                                                                                                                                                                                 |            |       |
| 8. (a)            | Find the minimum spanning tree of the following graph using Kruskal's Algorithm. Clearly show every step and calculate the total cost of the obtained MST:<br> | 5          | CO3   |
| (b)               | Write complete algorithms for Depth-First Search (DFS) and Breadth-First Search (BFS) graph traversals.                                                                                                                                         | 5          | CO3   |
| (c)               | Explain the following graph terminology with illustrative examples:<br>(i) Directed Graph (ii) Undirected Graph (iii) Path<br>(iv) Cycle (v) Degree of a vertex                                                                                 | 5          | CO3   |
| 9. (a)            | Consider the following graph and find out the shortest path from vertex 1 to all other vertices using Dijkstra's Algorithm:<br>                              | 5          | CO3   |
| (b)               | Perform DFS and BFS on the following graph considering vertex A as start vertex. Show the resultant spanning trees generated using DFS and BFS:<br>          | 5          | CO3   |
| (c)               | Write a short note on disjoint sets representation (Find and Union operations with path compression).                                                                                                                                           | 5          | CO3   |

\*\*\* End of Question Paper \*\*\*